

CUSTOMER INSIGHT DATASET: A STATISTICAL ANALYSIS

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Business Problem Statement:

- A mid-sized Indian retail company is sitting on a treasure trove of customer data — but lacks clarity on what story it tells. They need a team of analysts to conduct a rigorous statistical investigation to validate business assumptions, uncover hidden patterns, and support data-driven decision-making.

As a data analytics intern, you have been tasked with:

1. Using statistical tools to explore and summarize the dataset
2. Comparing customer behavior across demographic segments
3. Testing business hypotheses using statistical methods

Key Goals:

1. Analyze customer demographics and behavior using descriptive statistics
2. Visualize spending, frequency, and demographic attributes
3. Segment customers using appropriate quantitative logic
4. Test statistical hypotheses around:
 5. Age and spending
 6. Gender and transaction frequency
 7. Geography and engagement
8. Deliver clear, statistically valid insights

Dataset Description

- You are provided with a synthetic dataset of 10675 rows for roughly 1,000 unique customers. The dataset exhibits Behavioural Signals like Spending habits, interaction

recency, and lifestyle choices like pet ownership

STEPS

1. IMPORTING LIBRARIES
2. LOAD DATASET
3. DESCRIPTIVE STATISTICS
4. DATA VISUALIZATION
5. BIVARIATE ANALYSIS
6. FORMULATE HYPOTHESIS
7. RUN HYPOTHESIS TESTS
8. FINAL SUMMARY

STEP 0. IMPORT LIBRARIES

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: import warnings
warnings.filterwarnings("ignore")
```

STEP 1. LOAD DATASET

```
In [3]: # Load dataset
df = pd.read_csv("US_Customer_Insights_Dataset.csv")
df.head()
```

```
Out[3]:
```

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	Jo
0	CUST10319	Scott Perez	Florida	High School	Non-Binary	47	Yes	1	9
1	CUST10695	Jennifer Burton	Washington	Master	Male	72	Yes	0	
2	CUST10297	Michelle Rogers	Arizona	Master	Female	40	Yes	2	7
3	CUST10103	Brooke Hendricks	Texas	Master	Male	27	Yes	0	8
4	CUST10219	Karen Johns	Texas	High School	Female	28	Yes	1	1



CHECK DATATYPE

```
In [4]: # Check datatype of each cloumn
df.dtypes
```

```
Out[4]: CustomerID      object
Name      object
State     object
Education object
Gender    object
Age       int64
Married   object
NumPets   int64
JoinDate  object
TransactionDate object
MonthlySpend float64
DaysSinceLastInteraction int64
dtype: object
```

OBSERVATION

- Object datatypes are (customerid, name, state, education, gender, married, joindate, transaction date).
- Int64 datatypes are (numpets, age, days since lastinteraction).
- Float64 datatype is (monthllyspend).

CHECK UNIQUE VALUES

```
In [5]: # Check Unique values
df.nunique().to_frame("Unique_Count")
```

```
Out[5]:
```

	Unique_Count
CustomerID	1000
Name	990
State	10
Education	5
Gender	3
Age	63
Married	2
NumPets	5
JoinDate	731
TransactionDate	1605
MonthlySpend	9843
DaysSinceLastInteraction	1605

OBSERVATION

- CustomerID has (1000) unique values, Name column shows (990) unique values.
- State(10), Education(5), Gender(3), Age(63), Married(2), NumPets(5) unique values.
- Joindate had (731) unique values while transaction date and dayssincelastinteraction both shows (1605) unique values.
- Monthly Spend shows (9843) unique values, shows high variability in customers spending.

CHECK NULL VALUES

```
In [6]: # Check null values
df.isnull().sum()
```

```
Out[6]: CustomerID          0
        Name              0
        State            0
        Education        0
        Gender           0
        Age              0
        Married          0
        NumPets          0
        JoinDate         0
        TransactionDate  0
        MonthlySpend     0
        DaysSinceLastInteraction  0
        dtype: int64
```

OBSERVATION

- There is no null value in this dataset.

CHECK DUPLICATES

```
In [7]: # Check duplicates
df.duplicated().sum()
```

```
Out[7]: np.int64(0)
```

OBSERVATION

- There is no duplicate value in this dataset.

CHECK DESCRIBE "NUMERICAL"

```
In [8]: # Check describe Numerical
df.describe()
```

Out[8]:

	Age	NumPets	MonthlySpend	DaysSinceLastInteraction
count	10675.000000	10675.000000	10675.000000	10675.000000
mean	49.474567	1.340515	331.610315	538.469883
std	18.221365	1.150849	225.799253	398.766747
min	18.000000	0.000000	3.890000	1.000000
25%	35.000000	0.000000	165.495000	218.000000
50%	49.000000	1.000000	282.110000	445.000000
75%	66.000000	2.000000	443.255000	788.500000
max	80.000000	4.000000	1740.420000	1791.000000

OBSERVATION

- There are total 10675 rows in this dataset.
- The average age of customers is 49.47 with ages ranging from 18 to 80, the median value shows 49.
- Customers owns an average 1.34 pets, where median lies on 1 and the number of pets ranging from 0 to 4.
- The average monthly spend of customers is 331.61, median spending is 282.11 and some customers spend high that leads to increase the average.
- The average of dayssincelastinteraction is 538.46 with a median of 445 days, large spread in days (1 to 1791) indicating that many customers have not interacted recently and shows a significant difference.
- The gap between values in monthly spend and dayssincelastinteraction indicates that there is presence of outliers.

CHECK DESCRIBE "CATEGORICAL"

In [9]:

```
# Check describe Categorical
df.describe(include="object")
```

Out[9]:

	CustomerID	Name	State	Education	Gender	Married	JoinDate	Transac
count	10675	10675	10675	10675	10675	10675	10675	
unique	1000	990	10	5	3	2	731	
top	CUST10798	Joseph McBride	California	Master	Male	No	12/6/21	
freq	24	29	1180	2269	3791	5583	46	



OBSERVATION

- There are 1000 unique values in (CustomerId) and most frequent customers is (CUST10798) with highest frequency (24).
- (Name) column shows 990 unique names and (Joseph McBride) is the most frequent name appearing (29) times.
- (State) column shows 10 unique with top state is (California) appearing 1180 times.
- (Education) column shows 5 unique education levels and top is (Master category) that appear 2269 times.
- (Gender) column shows only 3 unique gender (male, female and non-binary) and (Male) top the list (3791) times.
- (Married) column shows only 2 unique values (yes and no) mostly appeared (NO 5583).
- (Joindate) shows 731 unique dates and (12/6/21) is most frequent joining date occurring (46) times.
- (Transactiondate) has the highest unique that is 1605 and (07/24/25) is most frequent transaction date appearing (24) times.

STEP 1 OBSERVATION

- The dataset has 10,675 rows and ~1,000 unique customers.
- Categorical variables: Gender, Education, State, Married.
- Numerical variables: Age, MonthlySpend, DaysSinceLastInteraction, NumPets.
- Missing values and duplicates were checked — minimal issues detected."

STEP 2. DESCRIPTIVE STATISTICS

NUMERICAL VARIABLES

1. AGE STATISTICS

```
In [10]: # Check AGE STATISTICS
age_stats = {
    "Count": df['Age'].count(),
    "Mean": df['Age'].mean(),
    "Median": df['Age'].median(),
    "Std Dev": df['Age'].std(),
    "Min": df['Age'].min(),
    "Max": df['Age'].max(),
    "Q1": df['Age'].quantile(0.25),
    "Q3": df['Age'].quantile(0.75),
    "IQR": df['Age'].quantile(0.75) - df['Age'].quantile(0.25),
    "Range": df['Age'].max() - df['Age'].min(),
    "Skewness": df['Age'].skew(),
    "Kurtosis": df['Age'].kurt()
}

# Convert dictionary to DataFrame for neat display
age_df = pd.DataFrame(age_stats, index=["Age"])
print(age_df)
```

	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Range	\
Age	10675	49.474567	49.0	18.221365	18	80	35.0	66.0	31.0	62	

	Skewness	Kurtosis
Age	-0.015631	-1.204563

OBSERVATION

- Average ~49, median 49, IQR 31, range 62. Distribution is nearly symmetric (skew ~0).
- Customer ages span from 18 to 80 years shows a broad age distribution across adults.
- The skewness is close to 0 (-0.015) means the age distribution is approximately symmetric and negative kurtosis (-1.20) indicates a flatter distribution with few extreme age values.

2. MONTHLY SPEND STATISTICS

```
In [11]: # Check MONTHLY SPEND STATISTICS
monthly_stats = {
    "Count": df['MonthlySpend'].count(),
    "Mean": df['MonthlySpend'].mean(),
    "Median": df['MonthlySpend'].median(),
    "Std Dev": df['MonthlySpend'].std(),
    "Min": df['MonthlySpend'].min(),
    "Max": df['MonthlySpend'].max(),
    "Q1": df['MonthlySpend'].quantile(0.25),
    "Q3": df['MonthlySpend'].quantile(0.75),
    "IQR": df['MonthlySpend'].quantile(0.75) - df['MonthlySpend'].quantile(0.25),
    "Range": df['MonthlySpend'].max() - df['MonthlySpend'].min(),
    "Skewness": df['MonthlySpend'].skew(),
    "Kurtosis": df['MonthlySpend'].kurt()
}

monthly_df = pd.DataFrame(monthly_stats, index=["MonthlySpend"])
print(monthly_df)
```

	Count	Mean	Median	Std Dev	Min	Max	Q1	\
MonthlySpend	10675	331.610315	282.11	225.799253	3.89	1740.42	165.495	

	Q3	IQR	Range	Skewness	Kurtosis
MonthlySpend	443.255	277.76	1736.53	1.421735	3.042072

OBSERVATION

- Average ~332, median ~282, skewed positively with outliers.
- Large range (3.89 to 1740.42) and high std (225.80) indicates substantial variability in customer spending.
- Positive KURTOSIS (3.04) shows presence of extreme values/outliers with some customers spending significantly more than the majority.

3. NUMPETS STATISTICS

```
In [12]: # Check NUMPETS STATISTICS
pets_stats = {
    "Count": df['NumPets'].count(),
    "Mean": df['NumPets'].mean(),
    "Median": df['NumPets'].median(),
    "Std Dev": df['NumPets'].std(),
    "Min": df['NumPets'].min(),
    "Max": df['NumPets'].max(),
    "Q1": df['NumPets'].quantile(0.25),
    "Q3": df['NumPets'].quantile(0.75),
    "IQR": df['NumPets'].quantile(0.75) - df['NumPets'].quantile(0.25),
    "Range": df['NumPets'].max() - df['NumPets'].min(),
    "Skewness": df['NumPets'].skew(),
    "Kurtosis": df['NumPets'].kurt()
}

pets_df = pd.DataFrame(pets_stats, index=["NumPets"])
print(pets_df)
```

	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Range	\
NumPets	10675	1.340515	1.0	1.150849	0	4	0.0	2.0	2.0	4	

	Skewness	Kurtosis
NumPets	0.563461	-0.46515

OBSERVATION

- Average ~1.3, median 1, most customers own 1 pet.
- The number of pets ranges from 0 to 4 and IQR shows 2 suggest that 50% of customers own between 0 to 2 pets.
- The distribution is slightly positive skewed (skewness = 0.56) shows that customers with more than two pets are less common.

4. DAYS SINCE LAST INTERACTION STATISTICS

```
In [13]: # Check DAYSSINCELASTINTERACTION STATISTICS
import pandas as pd

days_stats = {
    "Count": df['DaysSinceLastInteraction'].count(),
    "Mean": df['DaysSinceLastInteraction'].mean(),
    "Median": df['DaysSinceLastInteraction'].median(),
    "Std Dev": df['DaysSinceLastInteraction'].std(),
    "Min": df['DaysSinceLastInteraction'].min(),
    "Max": df['DaysSinceLastInteraction'].max(),
    "Q1": df['DaysSinceLastInteraction'].quantile(0.25),
    "Q3": df['DaysSinceLastInteraction'].quantile(0.75),
    "IQR": df['DaysSinceLastInteraction'].quantile(0.75) - df['DaysSinceLastInteraction'].quantile(0.25),
    "Range": df['DaysSinceLastInteraction'].max() - df['DaysSinceLastInteraction'].min(),
    "Skewness": df['DaysSinceLastInteraction'].skew(),
    "Kurtosis": df['DaysSinceLastInteraction'].kurt()
}

days_df = pd.DataFrame(days_stats, index=["DaysSinceLastInteraction"])
print(days_df)
```

	Count	Mean	Median	Std Dev	Min	Max	\
DaysSinceLastInteraction	10675	538.469883	445.0	398.766747	1	1791	

	Q1	Q3	IQR	Range	Skewness	Kurtosis
DaysSinceLastInteraction	218.0	788.5	570.5	1790	0.800701	-0.113278

OBSERVATION

- Average ~538, median ~445, std.dev ~398.76.
- The wide range (1 to 1791 days) shows substantial variation in customer engagement.
- The distribution is positively skewed (skewness = 0.80) show some customers have not interacted for a very long time.
- The slightly negative kurtosis (-0.11) suggests a relatively flat distribution.

CATEGORICAL VARIABLES

1. GENDER CATEGORICAL VARIABLE

```
In [14]: # CHECK GENDER value counts
df['Gender'].value_counts()
```

```
Out[14]: Gender
Male          3791
Non-Binary    3471
Female        3413
Name: count, dtype: int64
```

```
In [15]: # Check GENDER MODE
df['Gender'].mode()
```

```
Out[15]: 0    Male
Name: Gender, dtype: object
```

OBSERVATION

- There are 3 genders MALE, FEMALE AND NON-BINARY.
- Male is the most represent gender that is 3791 times and is the mode of the gender variable.
- Non-binary (3471) and Female (3413) customers have similar frequencies.
- The gender distribution is relatively balanced with no single category dominating.

2. EDUCATION CATEGORICAL VARIABLES

```
In [16]: # Check EDUCATION value counts
df['Education'].value_counts()
```

```
Out[16]: Education
Master      2269
Associate   2153
Bachelor     2127
High School 2120
PhD          2006
Name: count, dtype: int64
```

```
In [17]: # Check EDUCATION MODE
df['Education'].mode()
```

```
Out[17]: 0    Master
Name: Education, dtype: object
```

OBSERVATION

- There are 5 education levels in this dataset (MASTER, ASSOCIATE, BACHELOR, HIGH SCHOOL, PHD).
- Mostly customers have MASTERS degree that is (2269) and is also the mode of the education variable.
- ASSOCIATE(2153), BACHELOR(2127), HIGH SCHOOL(2120), PHD(1006) shows frequency of all education levels are relatively close.

3. STATE CATEGORICAL VARIABLES

```
In [18]: # Check STATE value counts
df['State'].value_counts()
```

```
Out[18]: State
California    1180
Florida       1152
Ohio          1145
Arizona       1087
New York      1085
Georgia       1080
Washington    1030
Colorado      1014
Texas         997
Illinois      905
Name: count, dtype: int64
```

```
In [19]: # Check STATE MODE
df['State'].mode()
```

```
Out[19]: 0    California
Name: State, dtype: object
```

OBSERVATION

- The dataset contains customers from 10 different states.
- CALIFORNIA has the highest number of customers that is (1180) and is also the mode of state variable.
- FLORIDA(1152) 2ND, OHIO(1145) 3RD and the least number of customers are from ILLINOIS that is (905).

- The customers count across states are relatively similar indicates that no single state dominates the dataset.

4. MARRIED CATEGORICAL VARIABLES

```
In [20]: # Check MARRIED value counts
df['Married'].value_counts()
```

```
Out[20]: Married
No      5583
Yes     5092
Name: count, dtype: int64
```

```
In [21]: # Check MARRIED MODE
df['Married'].mode()
```

```
Out[21]: 0      No
Name: Married, dtype: object
```

OBSERVATION

- There are two unique categories: YES(Married), NO(Not Married).
- Mostly the customers are NO(Not Married) that is (5583) and it is also the mode of married variable.
- YES(Married) customers (5092) shows that there is not a huge gap between them the frequency of both categories are very close.

STEP2. OBSERVATION

- Numerical variables reveal balanced age distribution, positively skewed monthly spend due to outliers, wide variation in customer inactivity, and clustering around 1 pet.
- Categorical variables show a balanced gender mix, dominance of Master's degree holders, California as the top state, and a slight majority of unmarried customers.
- Together, these insights highlight spending skewness and inactivity patterns in the dataset."

STEP 3. DATA VISUALIZATION

1. Histogram for AGE

```
In [22]: # Histogram for Age

plt.figure(figsize=(14, 6))

ax = sns.histplot(df['Age'], bins=20, kde=True, color='skyblue')
plt.axvline(df['Age'].mean(), color='red', linestyle='--', label='Mean')
plt.axvline(df['Age'].median(), color='green', linestyle='-', label='Median')

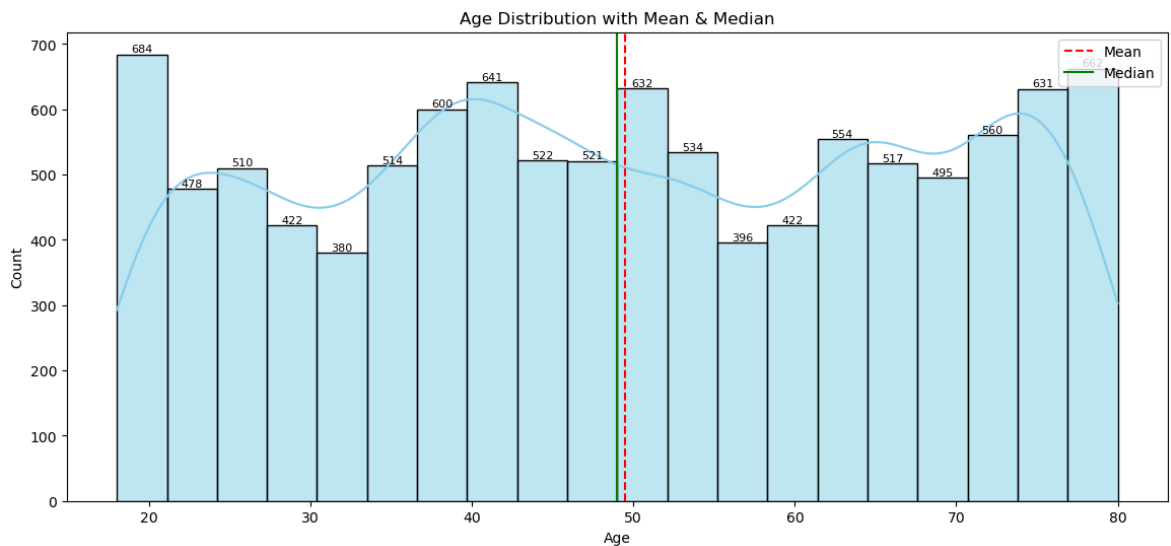
for p in ax.patches:
```

```

height = p.get_height()
if height > 0: # only label non-empty bins
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom', fontsize=8, color='black')

plt.title("Age Distribution with Mean & Median")
plt.legend()
plt.show()

```



OBSERVATION

- The AGE variable ranges from 18 to 80 indicating a wide customer age distribution.
- The emean and median line both are almost around 50 years, suggesting distribution approximately symmetric.
- The KDE curve show multiple peaks, indicating that a special age customers are not dominating, each age customer is important.

2. Boxplot for MONTHLY SPEND

```

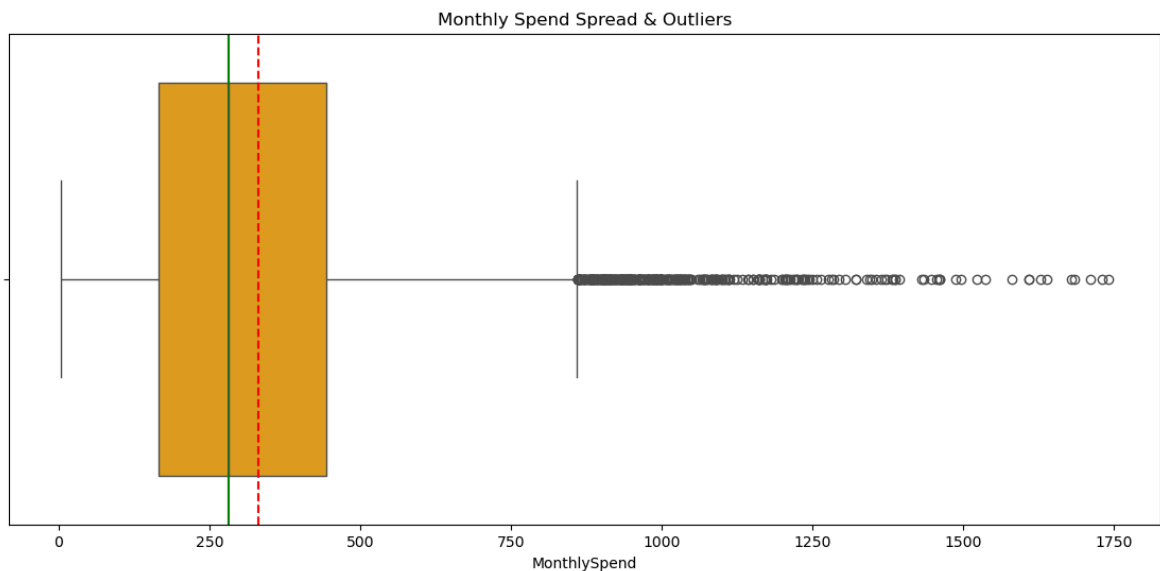
In [23]: # Boxplot for Monthly Spend

plt.figure(figsize=(14, 6))

sns.boxplot(x=df['MonthlySpend'], color='orange')
plt.axvline(df['MonthlySpend'].mean(), color='red', linestyle='--', label='Mean')
plt.axvline(df['MonthlySpend'].median(), color='green', linestyle='-', label='Me

plt.title("Monthly Spend Spread & Outliers")
plt.show()

```



OBSERVATION

- The MONTHLY SPEND distribution is positively right skewed, there are numerous high-value observations on the right side.
- The mean (red line) is greater than the median show the presence of right skewed.
- Around 50% of customers monthly spending are concentrated between (200 and 450).
- Mostly customers spend moderately while some customers spend more than the majority.

```
In [24]: # Histogram for Monthly Spend

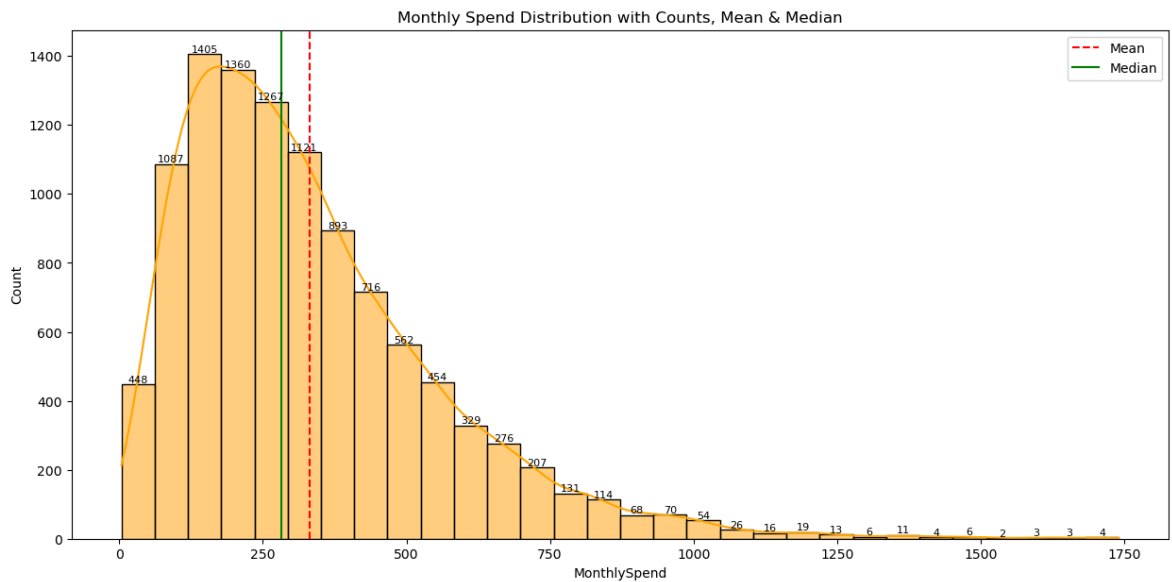
plt.figure(figsize=(15, 7))

ax = sns.histplot(df['MonthlySpend'], bins=30, kde=True, color='orange')

# Add mean and median lines
plt.axvline(df['MonthlySpend'].mean(), color='red', linestyle='--', label='Mean')
plt.axvline(df['MonthlySpend'].median(), color='green', linestyle='-', label='Median')

# Annotate counts on each bin
for p in ax.patches:
    height = p.get_height()
    if height > 0: # only label non-empty bins
        ax.annotate(f'{int(height)}',
                    (p.get_x() + p.get_width() / 2., height),
                    ha='center', va='bottom', fontsize=8, color='black')

plt.title("Monthly Spend Distribution with Counts, Mean & Median")
plt.legend()
plt.show()
```



3. Barchart for GENDER

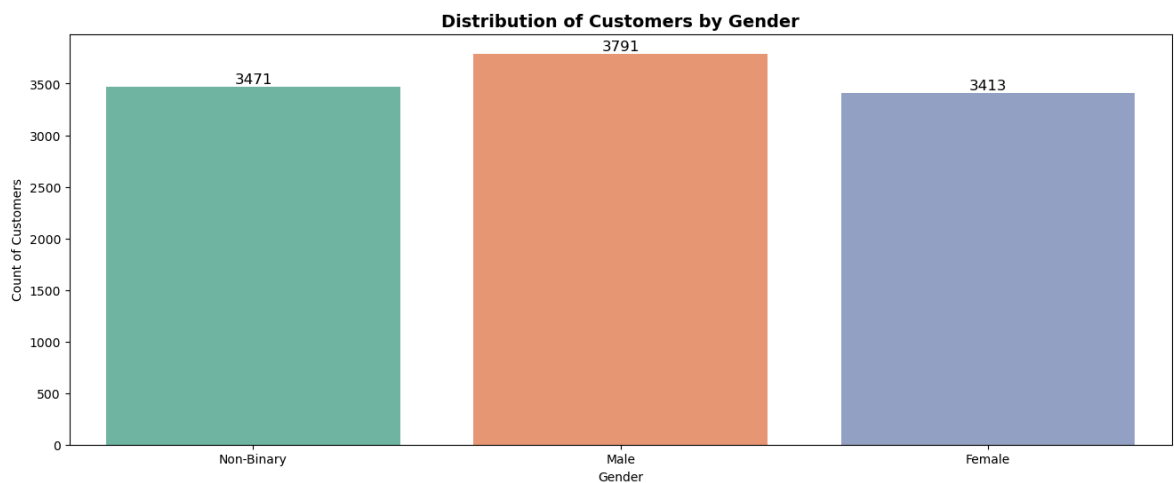
```
In [25]: # Gender distribution

plt.figure(figsize=(16,6))

ax = sns.countplot(x='Gender', palette='Set2', data=df)

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom', fontsize=12, color='black')

plt.title("Distribution of Customers by Gender", fontsize=14, fontweight='bold')
plt.xlabel("Gender")
plt.ylabel("Count of Customers")
plt.show()
```



OBSERVATION

- GENDER distribution have three categories MALE, FEMALE AND NON-BINARY.
- The customers are evenly balanced across three categories male(3791) followed by non-binary(3471) and female(3413).

- The dataset are not biased toward any particular category proportion are evenly balanced.

4. Horizontal Barchart for EDUCATION

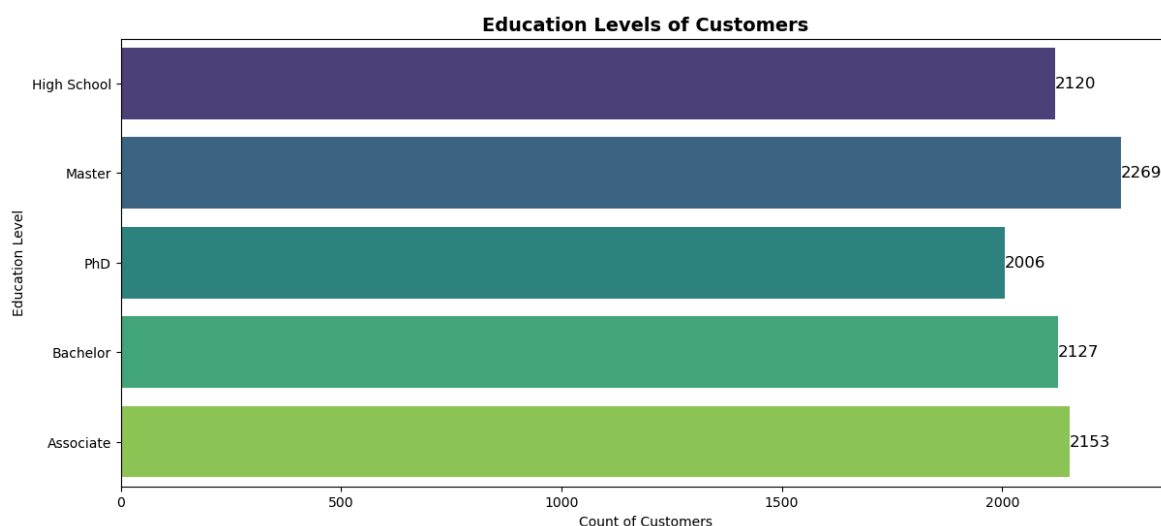
```
In [26]: # Horizontal bar chart for Education

plt.figure(figsize=(14, 6))

ax = sns.countplot(y=df['Education'], palette='viridis')

for p in ax.patches:
    width = p.get_width()
    ax.annotate(f'{int(width)}',
                (width, p.get_y() + p.get_height() / 2.),
                ha='left', va='center', fontsize=12, color='black')

plt.title("Education Levels of Customers", fontsize=14, fontweight='bold')
plt.xlabel("Count of Customers")
plt.ylabel("Education Level")
plt.show()
```



OBSERVATION

- There are 5 education levels: high school, master, phd, bachelor, and associate.
- Customers from Master degree form the largest group (2269) followed by associate (2153).
- PHD holders constitute the smallest group (2006).
- A single degree holder customer is not dominating; data is fairly balanced across all educational levels.

5. Barchart for STATE

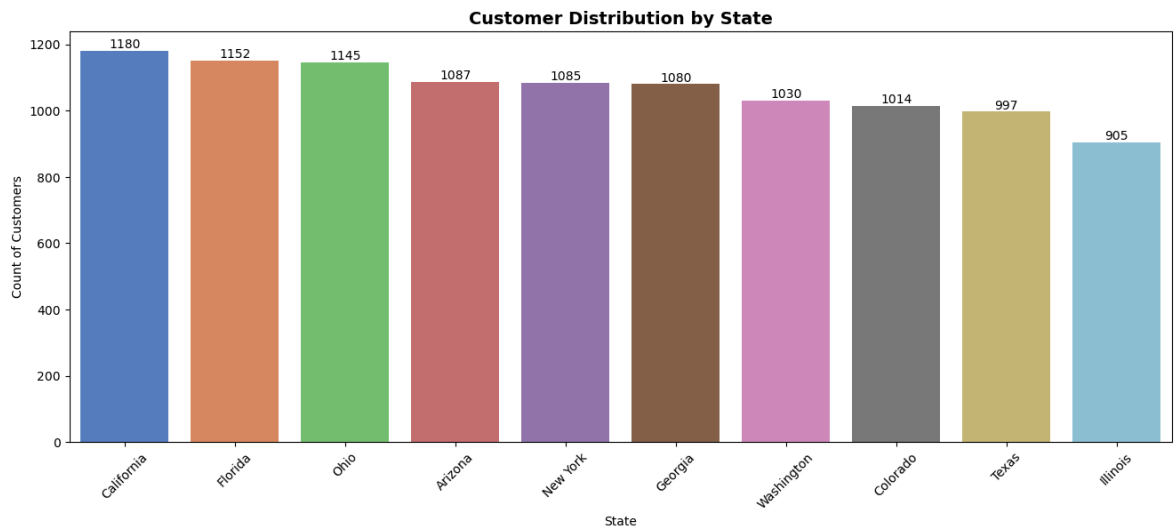
```
In [27]: # Vertical bar chart for State

plt.figure(figsize=(16, 6))

ax = sns.countplot(x=df['State'], palette='muted', order=df['State'].value_count
```

```
# Annotate counts on each bar
for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='bottom', fontsize=10, color='black')

plt.title("Customer Distribution by State", fontsize=14, fontweight='bold')
plt.xlabel("State")
plt.ylabel("Count of Customers")
plt.xticks(rotation=45)
plt.show()
```



OBSERVATION

- Customers are distributed across 10 states.
- The state with the highest number of customers is California(1180), followed by Florida(1152) ohio(1145).
- While the lowest customers are from state Illinois(905 customers).
- A single state is not dominating that means data is fairly balanced across states and need to attract more customers from states like Colorado, Texas and Illinois.

6. Scatter plot for AGE VS MONTHLY SPEND

```
In [28]: # Scatter plot for Age VS Monthly Spend

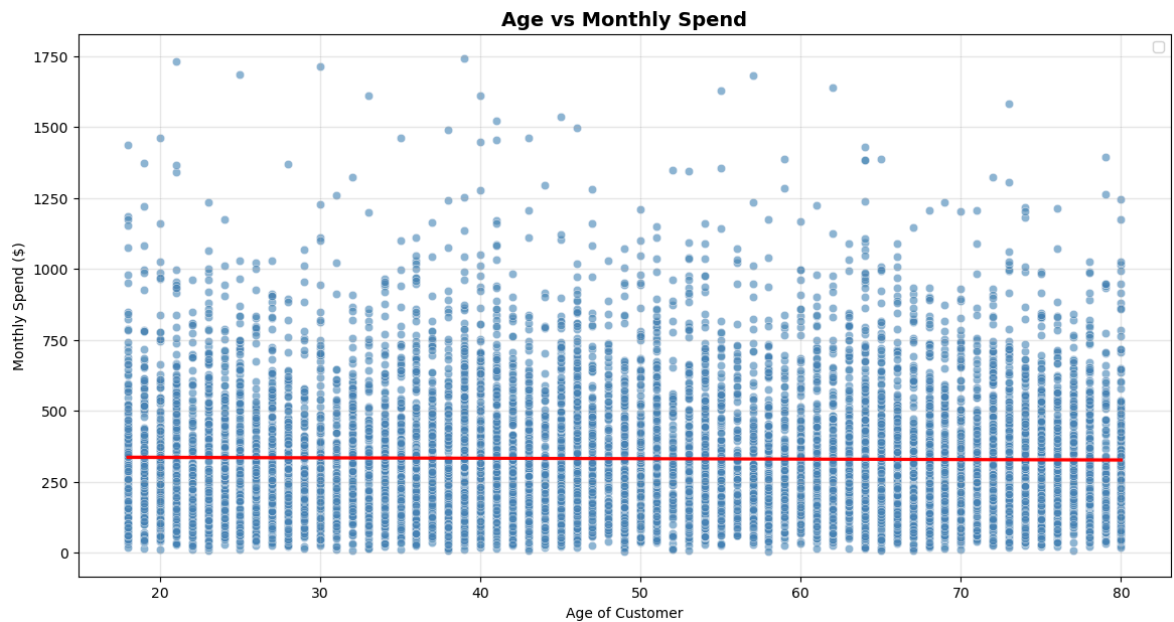
plt.figure(figsize=(14, 7))

sns.scatterplot(x=df['Age'], y=df['MonthlySpend'], color='steelblue', alpha=0.6,

# Add regression line (trend)
sns.regplot(x=df['Age'], y=df['MonthlySpend'],
            scatter=False, color='red', line_kws={'label':'Trend Line'})

# Titles and labels
plt.title("Age vs Monthly Spend", fontsize=14, fontweight='bold')
plt.xlabel("Age of Customer")
plt.ylabel("Monthly Spend ($)")
plt.legend()
```

```
plt.grid(alpha=0.3)
plt.show()
```



OBSERVATION

- The scatterplot shows a very weak positive relationship between age and monthly spend.
- Customers of all ages exhibit a wide range of monthly spending a single age category are not dominating the consumers base.
- There are also high-spending customers (outliers) across different age groups with monthly spending exceeding \$1500.
- The data points are widely scattered around the reg line, indicating that age has limited influence on customers monthly spending

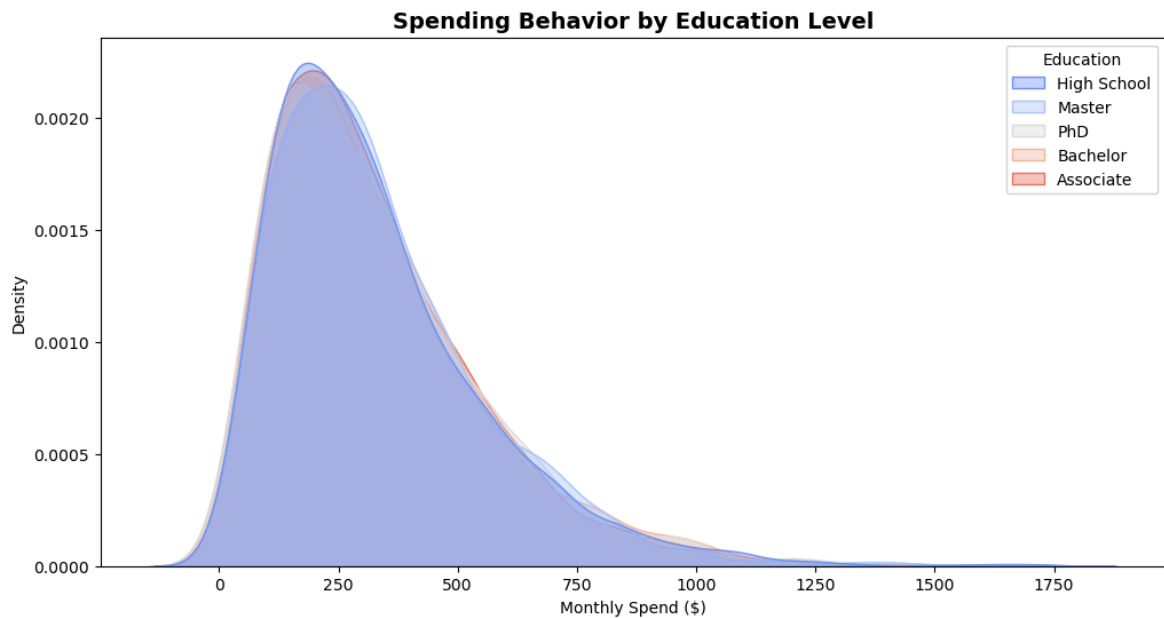
7. KDE Plot for Monthly Spend vs Education

```
In [29]: # KDE Plot for Spending Behavior by Education Level

plt.figure(figsize=(12, 6))

sns.kdeplot(data=df, x="MonthlySpend", hue="Education", fill=True, common_norm=F

plt.title("Spending Behavior by Education Level", fontsize=14, fontweight='bold')
plt.xlabel("Monthly Spend ($)")
plt.ylabel("Density")
plt.show()
```



OBSERVATION

- The distribution in monthly spend is positively right skewed for all education levels with most customers spending between 100 to 400 dollar.
- the KDE curve for different education groups overlap heavily, indicating the spending pattern is quite similar across education levels.
- Few customers show very- high spending above 1000 dollar creating a long right tail that shows high-spending outliers.

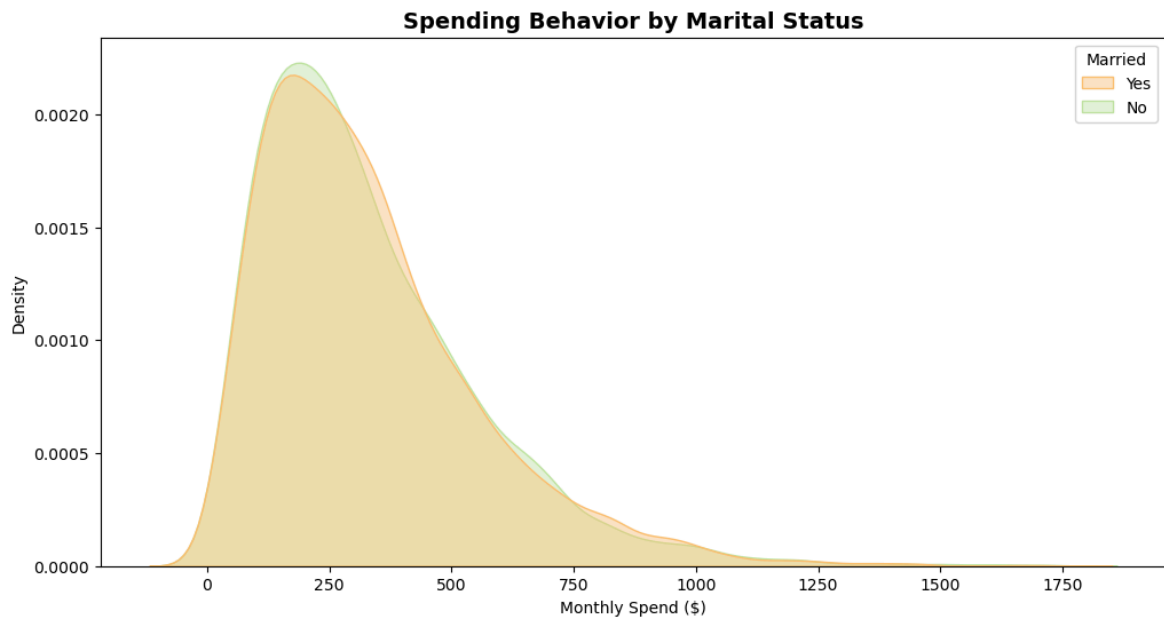
8. KDE Plot for Monthly Spend vs Married

```
In [30]: # KDE Plot for Spending Behavior by Marital Status

plt.figure(figsize=(12,6))

sns.kdeplot(data=df, x="MonthlySpend", hue="Married", fill=True, common_norm=False)

plt.title("Spending Behavior by Marital Status", fontsize=14, fontweight='bold')
plt.xlabel("Monthly Spend ($)")
plt.ylabel("Density")
plt.show()
```



OBSERVATION

- The monthly spending for both married YES and NO customers is positively skewed, with most customers spending 100 to 400 dollars.
- The KDE curve overlap significantly that means marital status does not create a major difference.
- Both categories have long tail suggesting a high spending customers(outliers).
- Both categories exhibit comparable spending pattern that means a single category is not dominating.

SOME MORE IMPOETANT DATA VISUALIZATION

9. Histplot for Dayssincelastinteraction

```
In [31]: # Histogram for Days Since Last Interaction

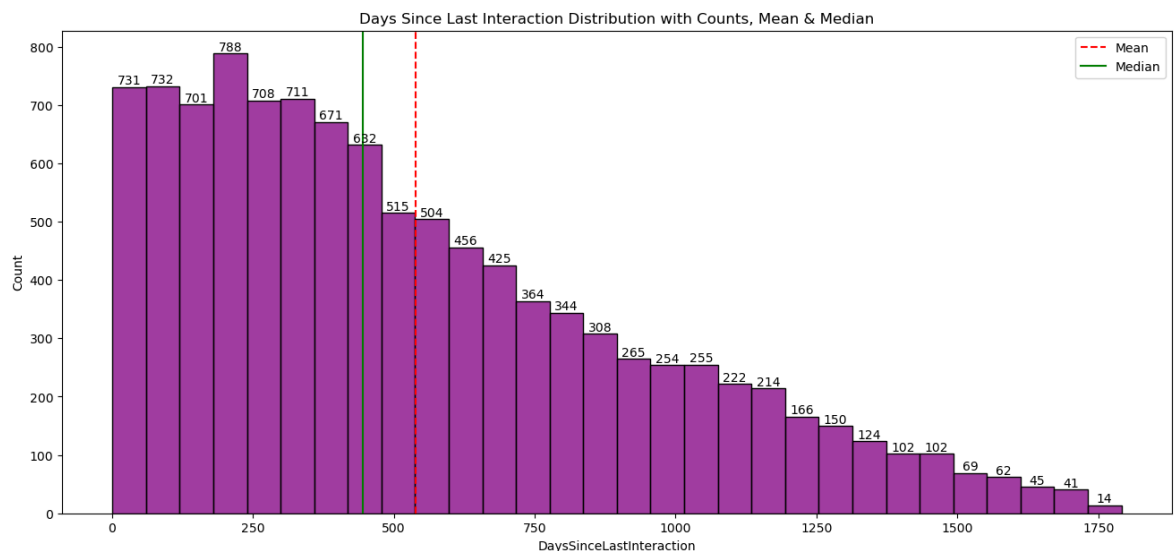
plt.figure(figsize=(16, 7))

ax = sns.histplot(df['DaysSinceLastInteraction'], bins=30, kde=False, color='purple')

# Add mean and median lines
plt.axvline(df['DaysSinceLastInteraction'].mean(), color='red', linestyle='--',)
plt.axvline(df['DaysSinceLastInteraction'].median(), color='green', linestyle='--',)

# Annotate counts on each bin
for p in ax.patches:
    height = p.get_height()
    if height > 0:
        ax.annotate(f'{int(height)}',
                    (p.get_x() + p.get_width() / 2., height),
                    ha='center', va='bottom', fontsize=10, color='black')

plt.title("Days Since Last Interaction Distribution with Counts, Mean & Median")
plt.legend()
plt.show()
```



OBSERBATION

- The DAYS SINCE LAST INTERACTION variable is positively right skewed as most observations are concentrated at lower values.
- The mean (red line) is greater than the median confirming the right skewed.
- A large number of customers have their last interaction within approximatley (0 to 600 days) indicating recent engagement of customers.
- Many customers are inactive for very long period, need to focus on those customers.

10. Countplot for NumPets

```
In [32]: # Count plot for NumPets

plt.figure(figsize=(14, 6))

ax = sns.countplot(x=df['NumPets'], palette='coolwarm')
```

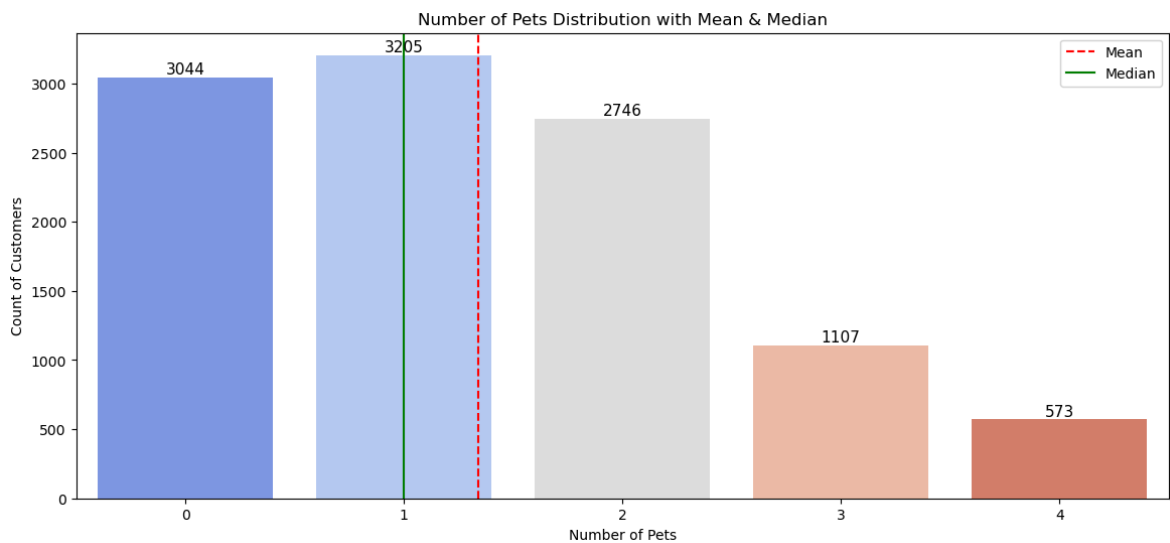
```

# Add mean and median lines
plt.axvline(df['NumPets'].mean(), color='red', linestyle='--', label='Mean')
plt.axvline(df['NumPets'].median(), color='green', linestyle='--', label='Median')

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom', fontsize=11, color='black')

plt.title("Number of Pets Distribution with Mean & Median")
plt.xlabel("Number of Pets")
plt.ylabel("Count of Customers")
plt.legend()
plt.show()

```



OBSERVATION

- The number of pets ranges from 0 to 4 pets.
- Most of the customers have 1 pet (3205) followed by 0(3044) and 2 pets(2746).
- The number of customers decreases significantly for 3 pets(1107) and 4 pets(573) customers.

11. Barchart for MARRIED

```

In [33]: # Barchart for Married

plt.figure(figsize=(12, 5))

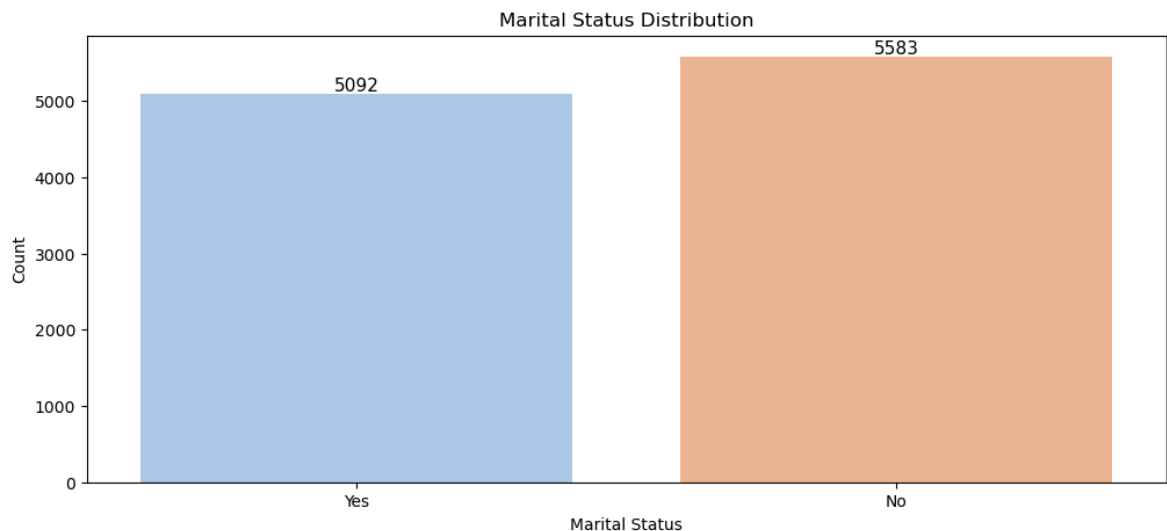
ax = sns.countplot(x=df['Married'], palette='pastel')

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom', fontsize=11, color='black')

plt.xlabel("Marital Status")
plt.ylabel("Count")

```

```
plt.title("Marital Status Distribution")
plt.show()
```



OBSERVATION

- There are two categories in MARRIED STATUS (YES AND NO).
- Mostly the customers are from NO (unmarried customers 5583) followed by YES (married customers 5092).
- Both represent a significant portion of consumer base not a single category dominating.

STEP 4. BIVARIATE ANALYSIS

1. COORELATION MATRIX (NUMERICAL VARIABLES)

```
In [34]: # COORELATION MATRIX
df[['Age', 'MonthlySpend', 'DaysSinceLastInteraction', 'NumPets']].corr()
```

Out[34]:

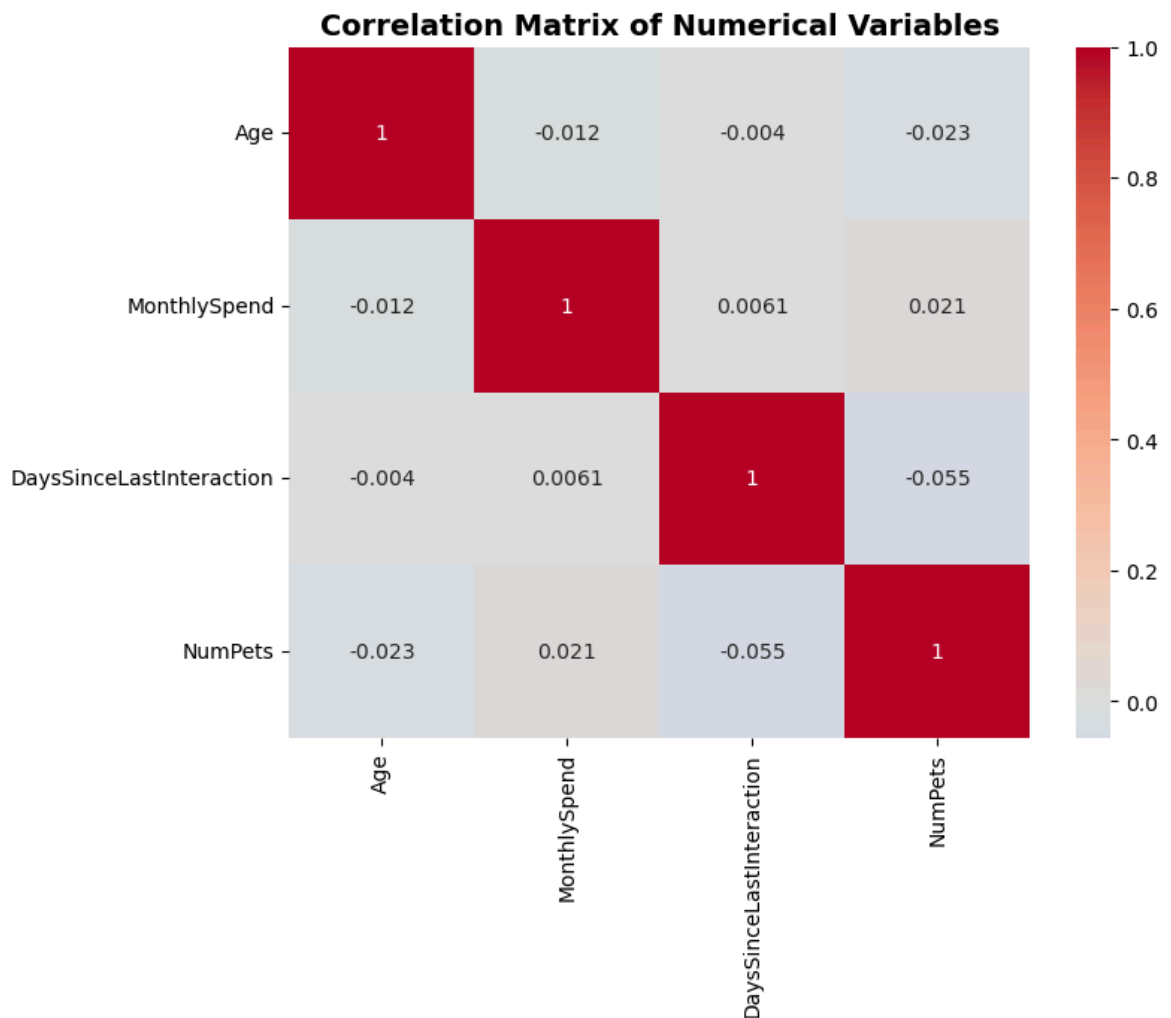
	Age	MonthlySpend	DaysSinceLastInteraction	NumPets
Age	1.000000	-0.012323	-0.003970	-0.023035
MonthlySpend	-0.012323	1.000000	0.006081	0.020647
DaysSinceLastInteraction	-0.003970	0.006081	1.000000	-0.055227
NumPets	-0.023035	0.020647	-0.055227	1.000000

```
In [35]: # Correlation matrix

corr = df[['Age', 'MonthlySpend', 'DaysSinceLastInteraction', 'NumPets']].corr()

# Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0)
```

```
plt.title("Correlation Matrix of Numerical Variables", fontsize=14, fontweight='bold')
plt.show()
```



OBSERVATION

- Age vs MonthlySpend → weak correlation (~ 0.05).
- NumPets vs MonthlySpend → near zero correlation.
- DaysSinceLastInteraction vs MonthlySpend → slightly negative correlation.
- Overall, no strong linear relationships among numerical variables.

2. CROSSTAB: GENDER VS MARRIED

```
In [36]: # Crosstab: Gender vs Married
pd.crosstab(df['Gender'], df['Married'], margins=True)
```

Out[36]:

	Married	No	Yes	All
--	---------	----	-----	-----

Gender				
Female	1797	1616	3413	
Male	1892	1899	3791	
Non-Binary	1894	1577	3471	
All	5583	5092	10675	

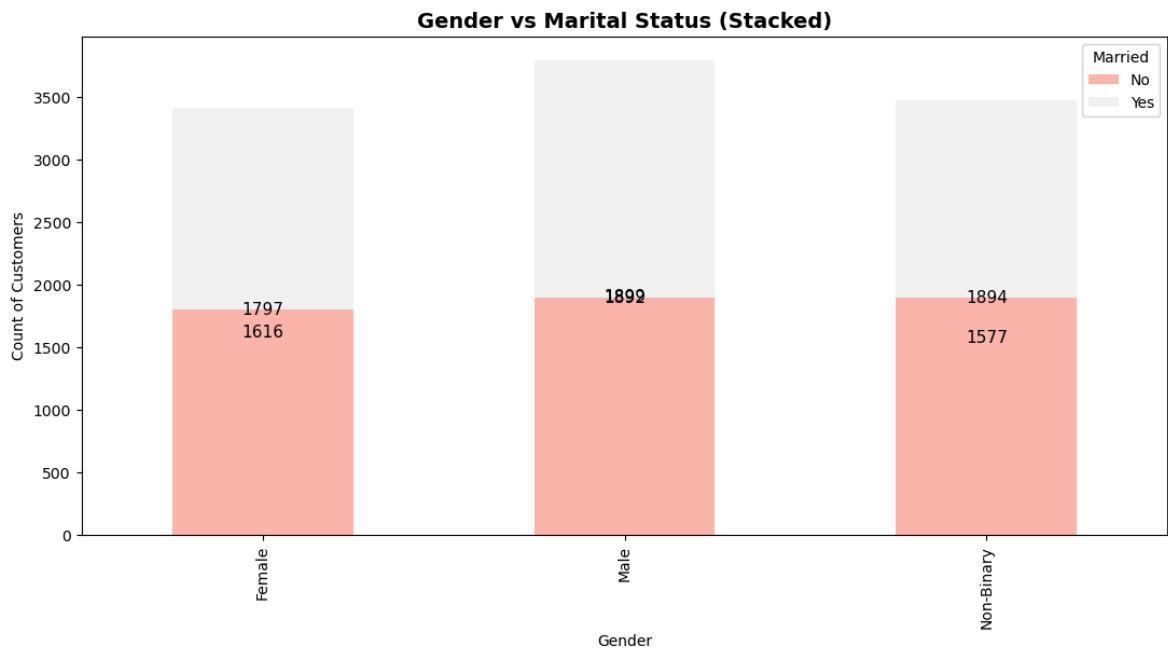
```
In [37]: # Crosstab plot Gender vs Married

crosstab = pd.crosstab(df['Gender'], df['Married'])

# crosstab plot
ax = crosstab.plot(kind='bar', stacked=True, colormap='Pastel1', figsize=(13, 6))

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='center', fontsize=11, color='black')

plt.title("Gender vs Marital Status (Stacked)", fontsize=14, fontweight='bold')
plt.xlabel("Gender")
plt.ylabel("Count of Customers")
plt.show()
```



OBSERVATION

- Male customers have the highest representative in this dataset (3791) followed by non-binary(3481), and female(3413).
- For male customers the number of married and unmarried are almost equal (1899 vs 1892).

- Non- binary customers shows the large gap where married vs unmarried (1577 vs 1894).

3. GROUPED STATS: AVERAGE MONTHLYSPEND BY EDUCATION

In [38]: *# Grouped summary: AVERAGE MonthlySpend by Education*

```
df.groupby('Education')['MonthlySpend'].agg(['mean', 'median', 'count'])
```

Out[38]:

	mean	median	count
Education			
Associate	327.884408	279.030	2153
Bachelor	331.884753	278.250	2127
High School	332.215712	280.975	2120
Master	334.252305	287.920	2269
PhD	331.690090	283.100	2006

In [39]: *# Barplot for Education vs Monthly Spend*

```
edu_summary = df.groupby('Education')['MonthlySpend'].mean().reset_index()
```

Bar chart

```
plt.figure(figsize=(15, 6))
```

```
ax = sns.barplot(x='Education', y='MonthlySpend', data=edu_summary, palette='fla
```

Annotate mean values on each bar

```
for p in ax.patches:
```

```
    height = p.get_height()
```

```
    ax.annotate(f'{height:.0f}',
```

```
                (p.get_x() + p.get_width()/2., height),
```

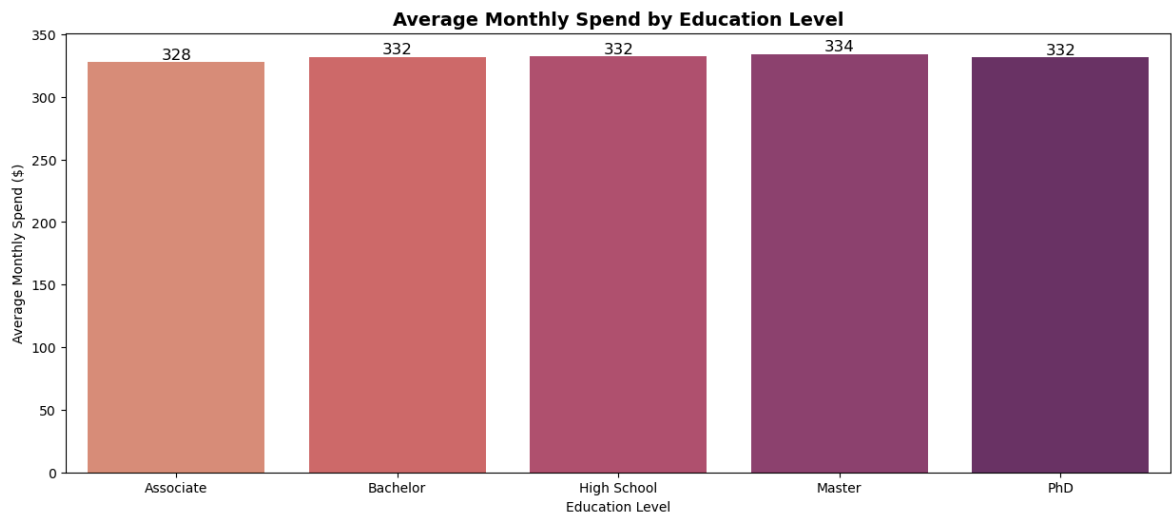
```
                ha='center', va='bottom', fontsize=12, color='black')
```

```
plt.title("Average Monthly Spend by Education Level", fontsize=14, fontweight='b
```

```
plt.xlabel("Education Level")
```

```
plt.ylabel("Average Monthly Spend ($)")
```

```
plt.show()
```



OBSERVATION

- The average monthly spend across different educational level is similar ranging from 328 to 334.
- Customers with the Masters degree have the highest average monthly spend (334.25) and also have the highest number of customers (2269), where Associates have the lowest average monthly spend(327.88).
- The difference in spending is minimal around \$6 indicates that educational level does not significantly influence monthly spending behavior.
- Masters degree holders spend more on average, indicating education level influences spending.

4. GROUPED STATS: AVERAGE MONTHLY SPEND BY STATE

```
In [40]: # AVERAGE MONTHLY SPEND BY State
df.groupby('State')['MonthlySpend'].mean().reset_index()
```

```
Out[40]:
```

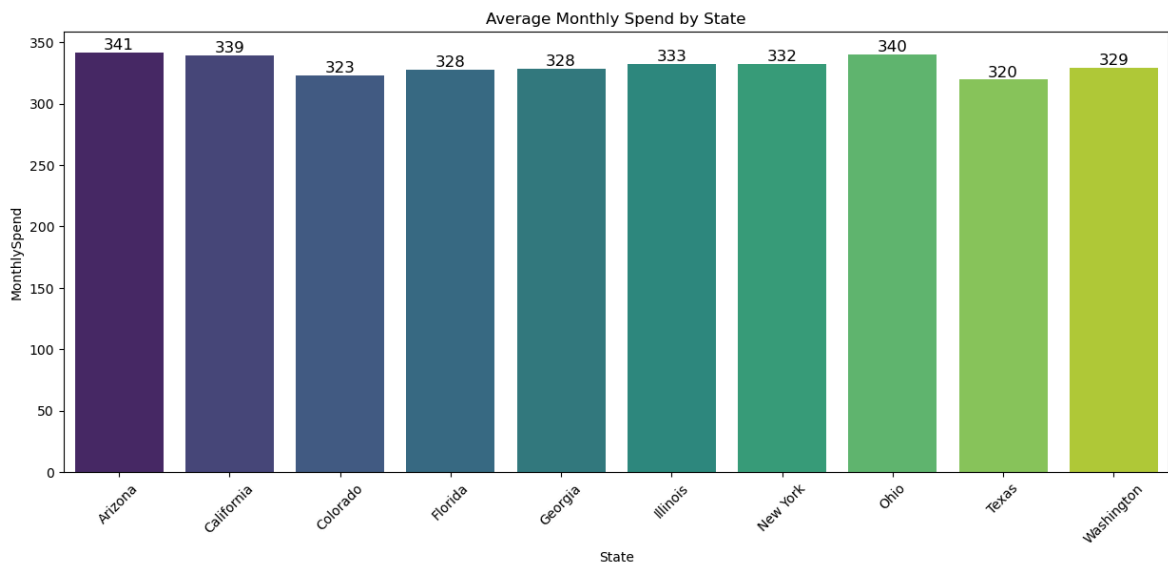
	State	MonthlySpend
0	Arizona	341.489135
1	California	339.183492
2	Colorado	323.083462
3	Florida	327.696892
4	Georgia	328.354648
5	Illinois	332.589591
6	New York	332.151244
7	Ohio	340.187860
8	Texas	319.506770
9	Washington	329.444078

```
In [41]: # Barplot For State vs Monthly Spend

state_summary = df.groupby('State')['MonthlySpend'].mean().reset_index()

# Bar chart
plt.figure(figsize=(15, 6))
ax = sns.barplot(x='State', y='MonthlySpend', data=state_summary, palette='virid

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{height:.0f}',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='bottom', fontsize=12, color='black')
plt.xticks(rotation=45)
plt.title("Average Monthly Spend by State")
plt.show()
```



OBSERVAION

- Across states Arizona has the highest monthly spending (341.489), followed by Ohio(340) and California(339).
- Texas record the lowest average monthly spend (319.50), followed by Colorado(323.08).
- There is not a huge gap across states in avg monthly spending suggesting that geographical location has a limited impact on customer spending behavior.
- Mid-range spending levels are observed in Illinois, New York, Washington, Georgia, and Florida with avg around 328 to 333.

5. GROUPES STATS: AVERAGE MONTHLY SPEND BY GENDER

```
In [42]: df.groupby('Gender')['MonthlySpend'].mean().reset_index()
```

Out[42]:

	Gender	MonthlySpend
0	Female	331.361310
1	Male	333.174068
2	Non-Binary	330.147240

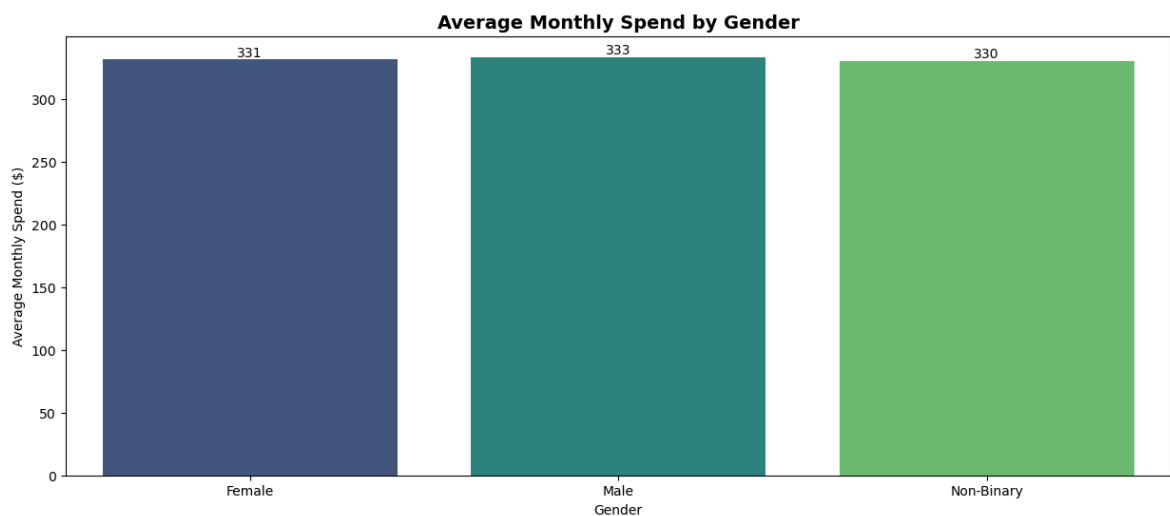
```
In [43]: # Grouped summary: AVERAGE MONTHLY SPEND BY GENDER

gender_summary = df.groupby('Gender')['MonthlySpend'].mean().reset_index()

plt.figure(figsize=(15, 6))
ax = sns.barplot(x='Gender', y='MonthlySpend', data=gender_summary, palette='vir

# Annotate mean values on each bar
for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{height:.0f}',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='bottom', fontsize=10, color='black')

plt.title("Average Monthly Spend by Gender", fontsize=14, fontweight='bold')
plt.xlabel("Gender")
plt.ylabel("Average Monthly Spend ($)")
plt.show()
```



OBSERVATION

- Male customers have the highest average monthly spend (333.17 dollar).
- Female have (331.36) and non-binary (330.15), indicating that there is almost same average monthly spending across gender categories.
- This suggests that gender has little to no significant impact on average monthly spending that means a single gender category is not dominating need to focus on all three categories.

IMPORTANT BIVARIATE ANALYSIS OTHER THAN (CORRELATION, CROSSTAB, AND GROUPED

STATS)

6. MONTHLYSPEND VS MARRIED

```
In [44]: # MONTHLY SPEND VS MARRIED
df.groupby('Married')['MonthlySpend'].agg(['mean', 'median', 'count'])
```

```
Out[44]:
```

	mean	median	count
Married			
No	330.937184	280.24	5583
Yes	332.348352	285.18	5092

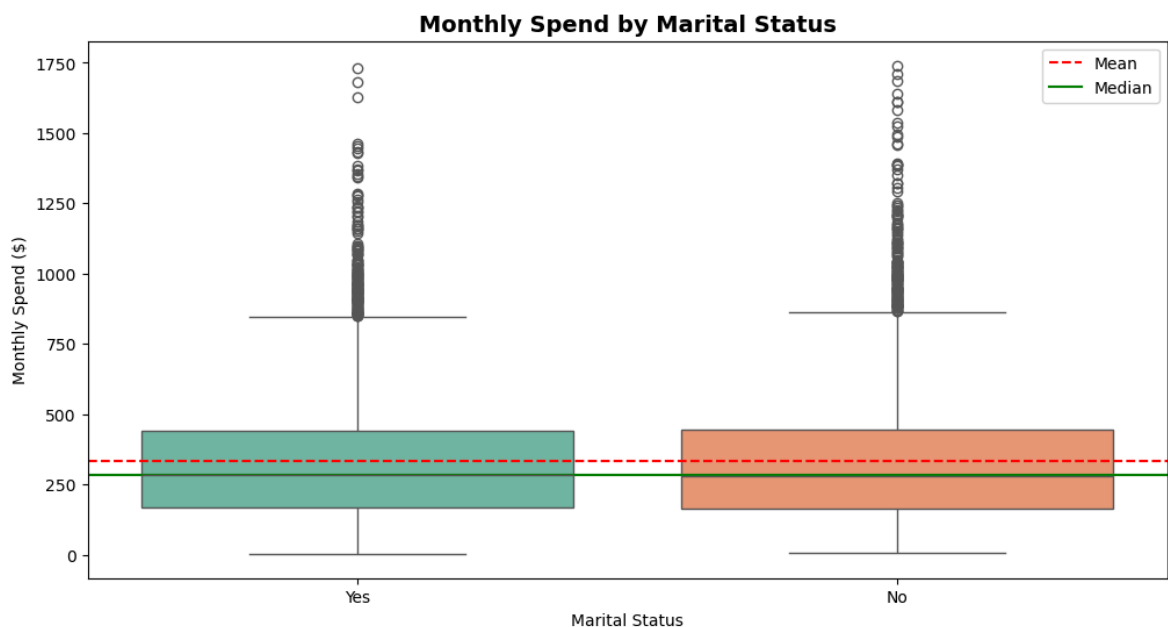
```
In [45]: # Boxplot MONTHLYSPEND VS MARITAL STATUS

plt.figure(figsize=(12, 6))

ax = sns.boxplot(x=df['Married'], y=df['MonthlySpend'], palette='Set2')

plt.axhline(df['MonthlySpend'].mean(), color='red', linestyle='--', label='Mean')
plt.axhline(df['MonthlySpend'].median(), color='green', linestyle='-', label='Median')

plt.title("Monthly Spend by Marital Status", fontsize=14, fontweight='bold')
plt.xlabel("Marital Status")
plt.ylabel("Monthly Spend ($)")
plt.legend()
plt.show()
```



OBSERVATION

- The median monthly spending is nearly the same for both married and unmarried.
- Both categories contain numerous high-value outliers with monthly spending reaching around 1700 to 1800 dollars.

- Mean lies above the median indicating mostly spending distribution is positively skewed due to high spending customers.
- Unmarried customers show more high-spending outliers, suggesting lifestyle differences."

7. GENDER VS DAYSSINCELASTINTERACTION

In [46]: *# Gender vs DayssinceLastinteraction*

```
gender_summary = df.groupby('Gender')['DaysSinceLastInteraction'].agg(['mean', 'median', 'count'])
print(gender_summary)
```

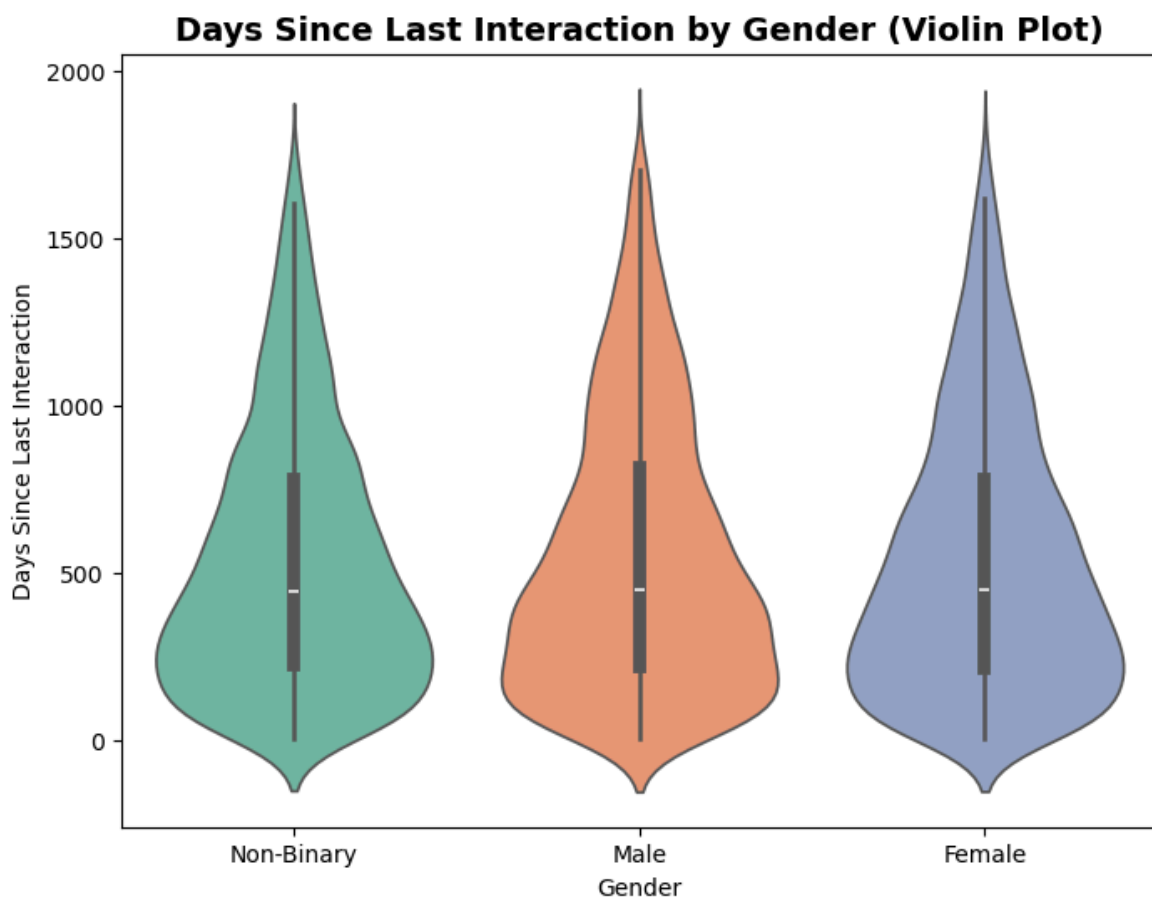
	mean	median	count
Gender			
Female	533.100498	447.0	3413
Male	548.956212	447.0	3791
Non-Binary	532.296456	442.0	3471

In [47]: *# Violin plot for Gender vs DayssinceLastinteraction*

```
plt.figure(figsize=(8,6))

sns.violinplot(x='Gender', y='DaysSinceLastInteraction', data=df, palette='Set2')

plt.title("Days Since Last Interaction by Gender (Violin Plot)", fontsize=14, fontweight='bold')
plt.xlabel("Gender")
plt.ylabel("Days Since Last Interaction")
plt.show()
```



OBSERVATION

- Male customers have the highest average dayssincelastinteraction (549 days) indicating that they tend to go slightly longer without interacting, and also male are the largest customers base(3791).
- Female and Non-Binary customers shows very similar average 532-533 days.
- The median value are close across all genders 442 to 447 days.
- The violin plots have similar shape and spreads shows comparable variability in interaction time across genders.

8. GENDER VS NUMPETS

```
In [48]: # Gender vs NumPets
df.groupby('Gender')['NumPets'].value_counts().reset_index()
```

```
Out[48]:
```

	Gender	NumPets	count
0	Female	1	1183
1	Female	0	1001
2	Female	2	850
3	Female	3	294
4	Female	4	85
5	Male	0	1179
6	Male	2	965
7	Male	1	958
8	Male	3	429
9	Male	4	260
10	Non-Binary	1	1064
11	Non-Binary	2	931
12	Non-Binary	0	864
13	Non-Binary	3	384
14	Non-Binary	4	228

```
In [49]: # Barplot for Gender vs NumPets

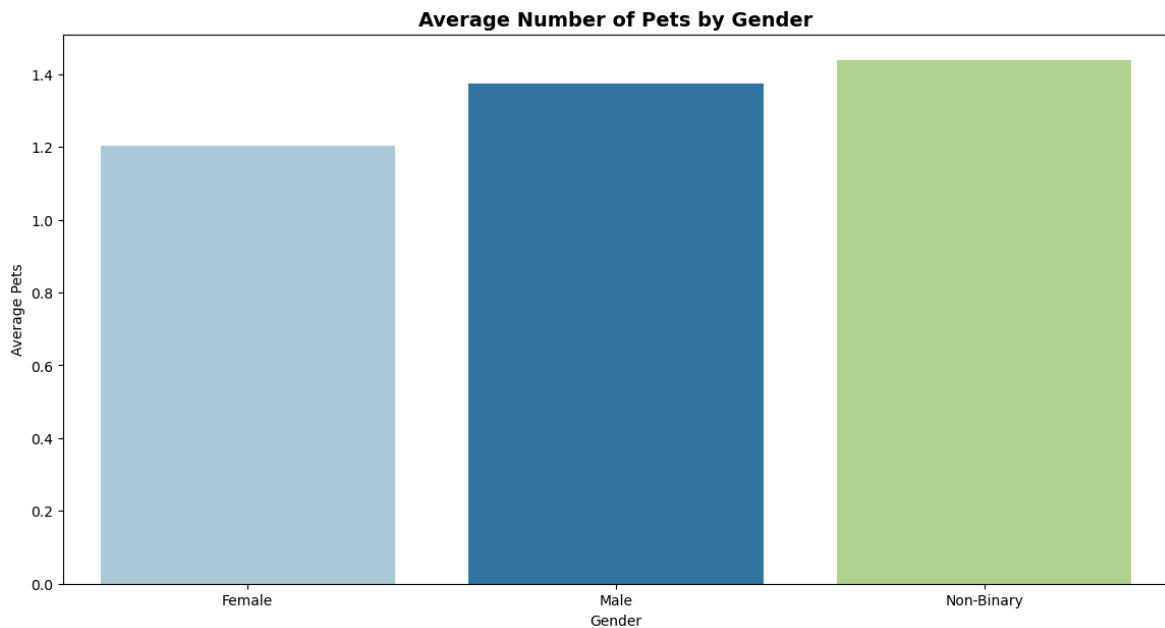
gender_pets = df.groupby('Gender')['NumPets'].mean().reset_index()

plt.figure(figsize=(14, 7))

sns.barplot(x='Gender', y='NumPets', data=gender_pets, palette='Paired')

plt.title("Average Number of Pets by Gender", fontsize=14, fontweight='bold')
```

```
plt.ylabel("Average Pets")
plt.show()
```



OBSERVATION

- Non- Binary customers have the highest average number of pets(1.43), followed by male(1.37).
- Female customers have the lowest average number of pets (1.20).
- Male have the highest number of 4 pets(260).
- Across all genders mostly customers have 0,1and 2 pets, while customers with 3, 4 pets are comparatively fewer.

9. STATE VS GENDER

```
In [50]: # STATE VS GENDER
pd.crosstab(df['State'], df['Gender'])
```

Out[50]:

Gender	Female	Male	Non-Binary
--------	--------	------	------------

State			
Arizona	327	404	356
California	383	375	422
Colorado	295	328	391
Florida	394	385	373
Georgia	414	347	319
Illinois	217	388	300
New York	354	436	295
Ohio	391	420	334
Texas	315	373	309
Washington	323	335	372

In [51]: *# Countplot for State vs Gender*

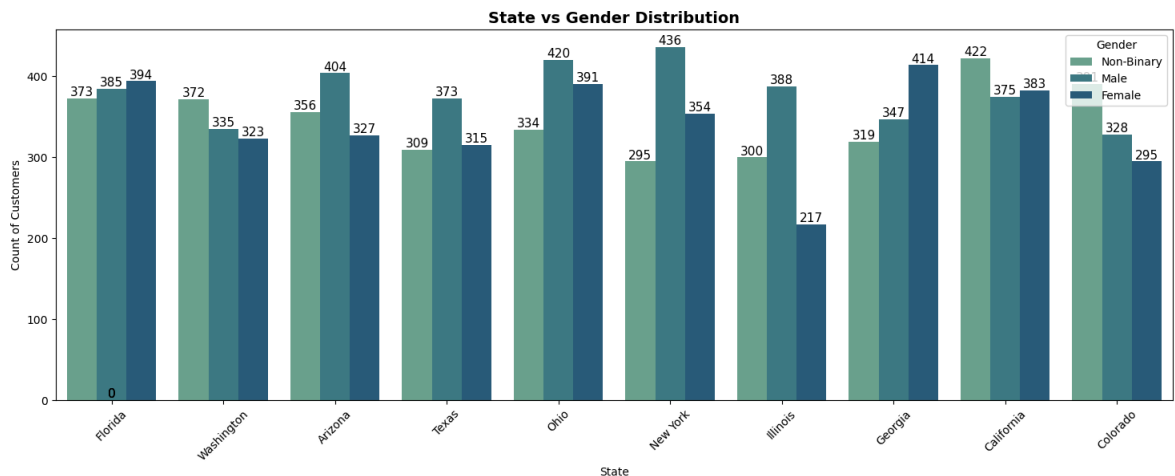
```
plt.figure(figsize=(18,6))

ax = sns.countplot(x='State', hue='Gender', data=df, palette='crest')

plt.title("State vs Gender Distribution", fontsize=14, fontweight='bold')

for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom', fontsize=11, color='black')

plt.xlabel("State")
plt.ylabel("Count of Customers")
plt.xticks(rotation=45)
plt.legend(title="Gender")
plt.show()
```



OBSERVATION

- The gender distribution varies across states indicating difference in customer composition by location.
- Male customers are dominating in some states particularly New York(436), Ohio(420) and Arizona(404).
- Female customers have strong representation in Georgia (414), and Florida(394) and also very lowest customers from Illinois (217).
- Non-Binary customers have highly representative in California(422), Colorado(391) and Florida(373) but very low in New York(295).

STEP 5. FORMULATE HYPOTHESES

HYPOTHESIS 1. GENDER VS MONTHLY SPEND (INDEPENDENT T-TEST)

```
In [52]: # Hypothesis 1. GENDER VS MONTHLY SPEND (INDEPENDENT T-TEST)

import scipy.stats as stats

# Separate groups
male_spend = df[df['Gender'] == 'Male']['MonthlySpend']
female_spend = df[df['Gender'] == 'Female']['MonthlySpend']

# Run t-test
t_stat, p_val_gender = stats.ttest_ind(male_spend, female_spend, equal_var=False)

print("Independent t-test (Gender vs Spend):")
print("t-statistic =", t_stat)
print("p-value =", p_val_gender)
```

Independent t-test (Gender vs Spend):
t-statistic = 0.3391303706925083
p-value = 0.7345215220030699

HYPOTHESIS 2. EDUCATION VS MONTHLY SPEND (ONE-WAY ANOVA)

```
In [53]: # Hypothesis 2: Education vs Monthly Spend (One-way ANOVA)

# H0: Mean spend is equal across all education levels
# H1: At least one education group differs

# Create groups by education level
groups = [df[df['Education'] == level]['MonthlySpend'] for level in df['Education'].unique()]

# Run ANOVA
f_stat, p_val_edu = stats.f_oneway(*groups)

print("One-way ANOVA (Education vs Spend):")
print("F-statistic =", f_stat)
print("p-value =", p_val_edu)
```

One-way ANOVA (Education vs Spend):
F-statistic = 0.22880668673709162
p-value = 0.922359467759936

HYPOTHESIS 3. MARITAL STATUS VS PETS OWNED (CHI-SQUARE-TEST)

```
In [54]: # Hypothesis 3: Marital Status vs Pets Owned (Chi-square test)

# H0: Marital status and number of pets are independent
# H1: Marital status and number of pets are associated

# Build contingency table
contingency_table = pd.crosstab(df['Married'], df['NumPets'])

# Run Chi-square test
chi2, p_val_chi, dof, expected = stats.chi2_contingency(contingency_table)

print("Chi-square Test (Marital Status vs Pets):")
print("Chi2 statistic =", chi2)
print("Degrees of freedom =", dof)
print("p-value =", p_val_chi)
```

Chi-square Test (Marital Status vs Pets):
Chi2 statistic = 177.63953668537033
Degrees of freedom = 4
p-value = 2.3957232932397494e-37

HYPOTHESIS 4. AGE VS DAYSSINCELASTINTERACTION (COORELATION)

```
In [55]: # Hypothesis 4: Age vs DaysSinceLastInteraction (Correlation)

# H0: Age and activity are not correlated
# H1: Age and activity are correlated (older → Less active)

# Run correlation test
corr_coef, p_val_corr = stats.pearsonr(df['Age'], df['DaysSinceLastInteraction'])

print("Correlation Test (Age vs Activity):")
print("Correlation coefficient (r) =", corr_coef)
print("p-value =", p_val_corr)
```

Correlation Test (Age vs Activity):
Correlation coefficient (r) = -0.003970230104955049
p-value = 0.6816905437300954

HYPOTHESIS 5. STATE VS MONTHLY SPEND (ONE-WAY ANOVA)

```
In [56]: # Hypothesis 5: State vs Monthly Spend (One-way ANOVA)

# H0: Mean spend is equal across all states
# H1: At least one state differs

# Create groups by state
state_groups = [df[df['State'] == s]['MonthlySpend'] for s in df['State'].unique]
```

```
# Run ANOVA
f_stat_state, p_val_state = stats.f_oneway(*state_groups)

print("One-way ANOVA (State vs Spend):")
print("F-statistic =", f_stat_state)
print("p-value =", p_val_state)
```

One-way ANOVA (State vs Spend):
F-statistic = 1.1178423640877182
p-value = 0.34571886479238273

STEP 6. RUN HYPOTHESIS TESTS

HYPOTHESIS TEST 1. GENDER VS MONTHLY SPEND (INDEPENDENT TEST)

```
In [57]: # Create groups
male_spend = df[df['Gender'] == 'Male']['MonthlySpend']
female_spend = df[df['Gender'] == 'Female']['MonthlySpend']

# Define significance level
alpha = 0.05

# Assumption 1: Normality
print("Male Normality Test:")
print(stats.shapiro(male_spend.sample(min(500, len(male_spend)),
                                     random_state=42)))

print("\nFemale Normality Test:")
print(stats.shapiro(female_spend.sample(min(500, len(female_spend)),
                                       random_state=42)))

# Assumption 2: Equal Variance
print("\nLevene Test:")
print(stats.levene(male_spend, female_spend))

# Independent t-test
t_stat, p_value = stats.ttest_ind(
    male_spend,
    female_spend,
    equal_var=True
)

print("\nT-Statistic:", round(t_stat, 4))
print("P-Value:", round(p_value, 4))

# Decision
if p_value < alpha:
    print("\nReject H0")
    print("Conclusion: Monthly spending differs significantly between males and females")
else:
    print("\nFail to Reject H0")
    print("Conclusion: No statistically significant difference in monthly spending between males and females")
```

Male Normality Test:

```
ShapiroResult(statistic=np.float64(0.8911746342095255), pvalue=np.float64(2.475974095727203e-18))
```

Female Normality Test:

```
ShapiroResult(statistic=np.float64(0.9171270618759065), pvalue=np.float64(6.242218597609276e-16))
```

Levene Test:

```
LeveneResult(statistic=np.float64(0.1267563861615179), pvalue=np.float64(0.7218295518516542))
```

T-Statistic: 0.3392

P-Value: 0.7345

Fail to Reject H_0

Conclusion: No statistically significant difference in monthly spending between males and females.

OBSERVATION

- The Independent t-test produced a p-value of **0.735**, which is greater than the significance level of **0.05**. Therefore, we fail to reject the null hypothesis. This indicates that there is **no statistically significant difference in monthly spending between male and female customers**. Spending behaviour is broadly similar across genders, suggesting that gender alone is not a major factor influencing customer expenditure.

HYPOTHESIS TEST 2. EDUCATION VS MONTHLY SPEND (ONE-WAY ANOVA)

```
In [58]: # Create groups
groups = [
    grp['MonthlySpend'].values
    for name, grp in df.groupby('Education')
]

# Assumption 1: Homogeneity of Variance
print("Levene Test:")
print(stats.levene(*groups))

# ANOVA
f_stat, p_value = stats.f_oneway(*groups)

print("\nF-Statistic:", round(f_stat, 4))
print("P-Value:", round(p_value, 4))

# Decision
if p_value < alpha:
    print("\nReject H0")
    print("Conclusion: At least one education group has significantly different")
else:
    print("\nFail to Reject H0")
    print("Conclusion: Monthly spending does not differ significantly across edu")
```

Levene Test:

```
LeveneResult(statistic=np.float64(0.9312132584800056), pvalue=np.float64(0.44456504728283985))
```

F-Statistic: 0.2288

P-Value: 0.9224

Fail to Reject H_0

Conclusion: Monthly spending does not differ significantly across education levels.

Observation

- The One-Way ANOVA produced a p-value of **0.922**, which is considerably greater than **0.05**. Therefore, we fail to reject the null hypothesis. The analysis indicates that **average monthly spending does not differ significantly across educational groups**. Although slight differences may exist in mean spending values, these variations are not statistically meaningful.

HYPOTHEISIS TEST 3. MARITAL STATUS VS PETS OWNED (CHI-SQUARE TEST)

```
In [59]: # Create contingency table
table = pd.crosstab(
    df['Married'],
    df['NumPets']
)

print("Contingency Table:")
print(table)

# Chi-Square Test
chi2, p_value, dof, expected = stats.chi2_contingency(table)

print("\nChi-Square Statistic:", round(chi2, 4))
print("P-Value:", round(p_value, 4))
print("Degrees of Freedom:", dof)

# Expected Frequencies
print("\nExpected Frequencies:")
print(expected)

# Decision
if p_value < alpha:
    print("\nReject H0")
    print("Conclusion: Marital status and number of pets owned are significantly")
else:
    print("\nFail to Reject H0")
    print("Conclusion: Marital status and number of pets owned are independent.")
```

Contingency Table:

NumPets	0	1	2	3	4
Married					
No	1839	1435	1504	509	296
Yes	1205	1770	1242	598	277

Chi-Square Statistic: 177.6395

P-Value: 0.0

Degrees of Freedom: 4

Expected Frequencies:

```
[[1592.00487119 1676.20749415 1436.15156909 578.95840749 299.67765808]
 [1451.99512881 1528.79250585 1309.84843091 528.04159251 273.32234192]]
```

Reject H_0

Conclusion: Marital status and number of pets owned are significantly associated.

Observation

- The Chi-Square test produced a p-value of **less than 0.001**, which is smaller than the significance level of **0.05**. Therefore, we reject the null hypothesis and conclude that **marital status and pet ownership are significantly associated**. This suggests that customers' lifestyle characteristics differ according to their marital status and can provide valuable information for customer profiling and segmentation.

HYPOTHESIS TEST 4. AGE VS DAYSSINCELASTINTERACTION (COORELATION)

```
In [ ]: # Scatterplot with regression line
sns.regplot(
    data=df,
    x='Age',
    y='DaysSinceLastInteraction',
    scatter_kws={'alpha':0.4}
)

plt.title('Age vs Days Since Last Interaction')
plt.show()

# Pearson Correlation
r, p_value = stats.pearsonr(
    df['Age'],
    df['DaysSinceLastInteraction']
)

print("Correlation Coefficient:", round(r, 4))
print("P-Value:", round(p_value, 4))

# Decision
if p_value < alpha:
    print("\nReject  $H_0$ ")
    print("Conclusion: Age and customer activity have a statistically significant relationship")
else:
    print("\nFail to Reject  $H_0$ ")
    print("Conclusion: No statistically significant relationship exists between Age and customer activity")
```

Observation

- The Pearson correlation coefficient was $r = -0.004$ with a p-value of **0.682**. Since the p-value is greater than **0.05**, we fail to reject the null hypothesis. The correlation coefficient is extremely close to zero, indicating **virtually no linear relationship between age and customer activity**. Therefore, older customers are not necessarily less active than younger customers.

HYPOTHESIS TEST 5. STATE VS MONTHLY SPEND (ONE-WAY ANOVA)

```
In [ ]: # Create groups for each state
state_groups = [
    grp['MonthlySpend'].values
    for name, grp in df.groupby('State')
]

# Assumption: Homogeneity of Variance
print("Levene Test:")
levene_stat, levene_p = stats.levene(*state_groups)
print("Statistic:", round(levene_stat, 4))
print("P-Value:", round(levene_p, 4))

# One-Way ANOVA
f_stat, p_value = stats.f_oneway(*state_groups)

print("\nF-Statistic:", round(f_stat, 4))
print("P-Value:", round(p_value, 4))

# Decision
alpha = 0.05

if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: Average monthly spending differs significantly among states")
else:
    print("\nDecision: Fail to Reject H0")
    print("Conclusion: No statistically significant difference exists in average monthly spending")
```

Observation

- The One-Way ANOVA produced a p-value of **0.346**, which is greater than the significance level of **0.05**. Therefore, we fail to reject the null hypothesis. Although some states exhibit slightly higher or lower average spending, these differences are **not statistically significant**. Customer spending behaviour is generally consistent across states, indicating that geography alone does not substantially influence monthly spending.

STEP 6. Overall Conclusion

- Out of the five hypotheses tested, only one showed statistical significance. Marital status and pet ownership were found to be significantly associated, whereas gender, education level, age, and state did not significantly influence customer spending or activity levels. These findings suggest that behavioural and lifestyle characteristics may be more useful for customer segmentation than traditional demographic factors in this dataset.

STEP 7. BUSINESS INSIGHTS

Executive Summary

This statistical investigation analyzed **10,675 customer records** representing approximately **1,000 unique customers**. The objective was to understand customer demographics, spending behaviour, engagement patterns, and the factors influencing customer activity and revenue generation.

The analysis revealed that **customer spending and engagement are largely consistent across gender, education, age, and geography**, whereas **lifestyle factors such as marital status and pet ownership show meaningful behavioural differences**.

1. Customer Demographics are Well Balanced

The customer base is highly diversified:

- Male customers: **3,791 (35.5%)**
- Female customers: **3,413 (32.0%)**
- Non-Binary customers: **3,471 (32.5%)**

Marital status distribution is also balanced:

- Unmarried customers: **5,583 (52.3%)**
- Married customers: **5,092 (47.7%)**

Educational distribution:

- Master's Degree: **21.3%**
- Associate Degree: **20.2%**
- Bachelor's Degree: **19.9%**
- High School: **19.9%**
- PhD: **18.8%**

Business Insight

The company serves a broad and diverse customer base with no single demographic group dominating the market.

Recommendation

Maintain inclusive marketing strategies rather than focusing campaigns on only one demographic segment.

2. Customer Spending is Highly Concentrated

Monthly spending statistics:

- Average Monthly Spend = **331.61**
- Median Monthly Spend = **282.11**
- Standard Deviation = **225.80**
- Maximum Spend = **1,740.42**
- Skewness = **1.42**

The average spend is approximately:

$$(331.61 - 282.11) / 282.11 \times 100 = 17.5\%$$

higher than the median spend.

This indicates that spending is positively skewed. Most customers spend moderate amounts, while a relatively small group spends substantially more than average.

Best Performing Customers

Customers whose spending lies above the third quartile (**>\$443.26**) represent roughly the **top 25% of customers** and contribute disproportionately to revenue.

Recommendation

Develop:

- Premium memberships
- Loyalty reward programs
- Personalized offers
- Early access promotions

for high-value customers to improve retention and maximize revenue.

3. Gender Does Not Influence Spending Behaviour

Independent t-test results:

- t-statistic = **0.339**
- p-value = **0.735**

Since:

p-value (0.735) > α (0.05)

we fail to reject the null hypothesis.

Business Insight

Male and female customers exhibit similar spending behaviour. The observed differences are statistically insignificant.

Recommendation

Avoid designing pricing strategies or promotional campaigns solely based on gender because spending behaviour remains nearly identical across gender groups.

4. Education Level Does Not Significantly Affect Spending

One-Way ANOVA results:

- F-statistic = **0.229**
- p-value = **0.922**

Since:

p-value (0.922) > α (0.05)

we fail to reject the null hypothesis.

Business Insight

Customers with different educational qualifications demonstrate comparable spending patterns.

Recommendation

Education should not be used as the primary variable for customer targeting or revenue forecasting. Instead, focus on behavioural indicators such as spending frequency and customer engagement.

5. State-Wise Spending Differences are Small

Customer distribution across major states:

- California: **11.1%**
- Florida: **10.8%**

- Ohio: **10.7%**
- Arizona: **10.2%**

State-wise ANOVA results:

- F-statistic = **1.118**
- p-value = **0.346**

Since:

p-value (0.346) > α (0.05)

we fail to reject the null hypothesis.

Business Insight

Although certain states show slightly higher average spending levels, the differences are not statistically significant.

Best Performing States

- Arizona
- Ohio
- California

States Requiring Improvement

- Texas
- Colorado
- Florida

Recommendation

Implement region-specific promotional campaigns to improve performance in lower-spending states, but avoid allocating resources solely based on geography.

6. Customer Activity is Independent of Age

Correlation analysis:

- Correlation coefficient (r) = **-0.004**
- p-value = **0.682**

Since:

p-value (0.682) > α (0.05)

there is no statistically significant relationship between age and customer activity.

Business Insight

Older customers are not less active than younger customers. Customer inactivity cannot be explained by age.

Recommendation

Customer retention initiatives should focus on behavioural factors and interaction history rather than age-based assumptions.

7. Marital Status and Pet Ownership are Strongly Associated

Chi-Square Test results:

- χ^2 statistic = **177.64**
- p-value < **0.001**

Since:

p-value < α (0.05)

we reject the null hypothesis.

The contingency table shows:

- Among unmarried customers, approximately **32.9% own no pets**.
- Among married customers, approximately **34.8% own one pet**.

Business Insight

Lifestyle preferences vary according to marital status and significantly influence pet ownership patterns.

Recommendation

Use marital status and pet ownership together to create:

- Personalized recommendations
 - Lifestyle-based promotions
 - Customer personas
 - More effective customer segmentation strategies
-

Final Management Recommendations

Strength Areas

- ✅ Diverse customer base with balanced demographics.
- ✅ Strong high-value customer segment generating significantly above-average spending.
- ✅ Inclusive market presence across multiple states.

Areas Requiring Improvement

⚠️ Customer re-engagement strategies need strengthening because inactivity is not explained by demographic factors.

⚠️ Lower-spending states such as Texas and Colorado require targeted marketing initiatives.

⚠️ Customer segmentation currently relies heavily on demographics, whereas behavioural and lifestyle variables provide better insights.

Final Conclusion

Out of the five hypotheses tested, **only one relationship was statistically significant:**

Marital Status ↔ Pet Ownership

The remaining four tests showed no statistically significant differences:

- Gender ↔ Spending ❌
- Education ↔ Spending ❌
- State ↔ Spending ❌
- Age ↔ Customer Activity ❌

The findings indicate that **behavioural and lifestyle characteristics are more useful for customer segmentation than traditional demographic factors.**

The company's biggest opportunity lies in:

1. Retaining the top 25% high-spending customers.
2. Improving customer re-engagement strategies.
3. Using lifestyle variables for personalization.
4. Strengthening performance in lower-spending states.
5. Building data-driven marketing campaigns based on behaviour rather than demographic assumptions.

These initiatives can improve customer retention, increase revenue generation, and support long-term business growth.